

Felipe Meneguitti Dias

Models, signals and software for healthcare.



I am Felipe Meneguitti Dias, an engineer and researcher with a PhD from the University of São Paulo. I work on machine learning, computer vision and biomedical signals, from model development to integration into healthcare systems.

Selected projects

01 Gorgona

I led the deployment of a platform that collects ECGs from computers at an urgent care unit, runs the model at InCor and sends alerts to the clinical coordination group on Telegram.

Project leadership and initial system development

02 ECG classification at InCor

I developed research on ECG image classification and a proposed integration with hospital infrastructure, presenting results in an emergency room web application.

Method development and first author of the IEEE Access paper

03 RPMS

I worked on data collection with prototype watches, study protocols and PPG algorithms for a remote patient monitoring system.

Data collection protocols, signal processing and algorithm research

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ECG classification integrated into urgent care

I led the deployment of a platform that collects ECGs from computers at an urgent care unit, runs the model at InCor and sends alerts to the clinical coordination group on Telegram.

42,037

processed records

2 min

ECG to alert in the case

41 min

before the first report

Dashboard query on October 3, 2026: all units, full period, Processed status, Current of Injury and Other classes. The count represents examination records, not unique patients.

Objective

Flag ECGs with suspected current of injury and notify the clinical coordination team earlier. Automated analysis runs alongside the telemedicine workflow. Patient assessment and care decisions remain with the medical team.

Previous workflow

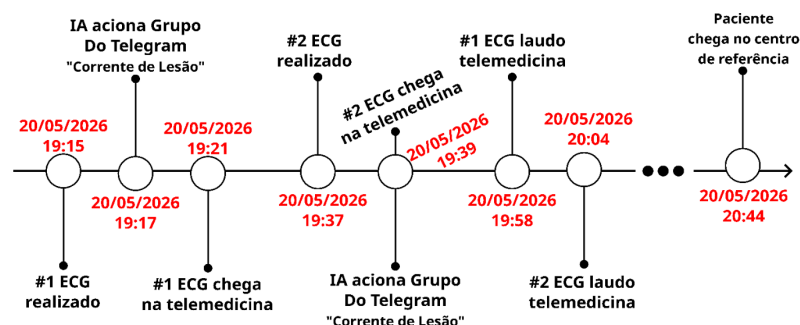
ECGs were sent to telemedicine. Reports and medical assessment informed contact with the coordination team.

With Gorgona

The Windows client detects the examination and sends it to InCor. The server classifies it and alerts the Telegram group when Current of Injury probability exceeds 50%.

I connected the client, server, web dashboard and alerts. The design includes configurable local anonymization, HTTPS and per-installation credentials.

Case study · UPA Rodrigo Argolo · 20/05/2026



Original timeline, slide 23. Personal identification block omitted.

Two ECGs from the same encounter triggered alerts within 2 minutes, preceding their reports by 41 and 25 minutes. The case documents communication timing.

Team publications

Image-based deep learning for emergency electrocardiogram classification. medRxiv · Preprint, 2026.

From PACS to the emergency room screen

I developed research on ECG image classification and a proposed integration with hospital infrastructure, presenting results in an emergency room web application.

Objective

Support the prioritization of ECG review in the emergency room by bringing automated classification into the clinical team's workspace.

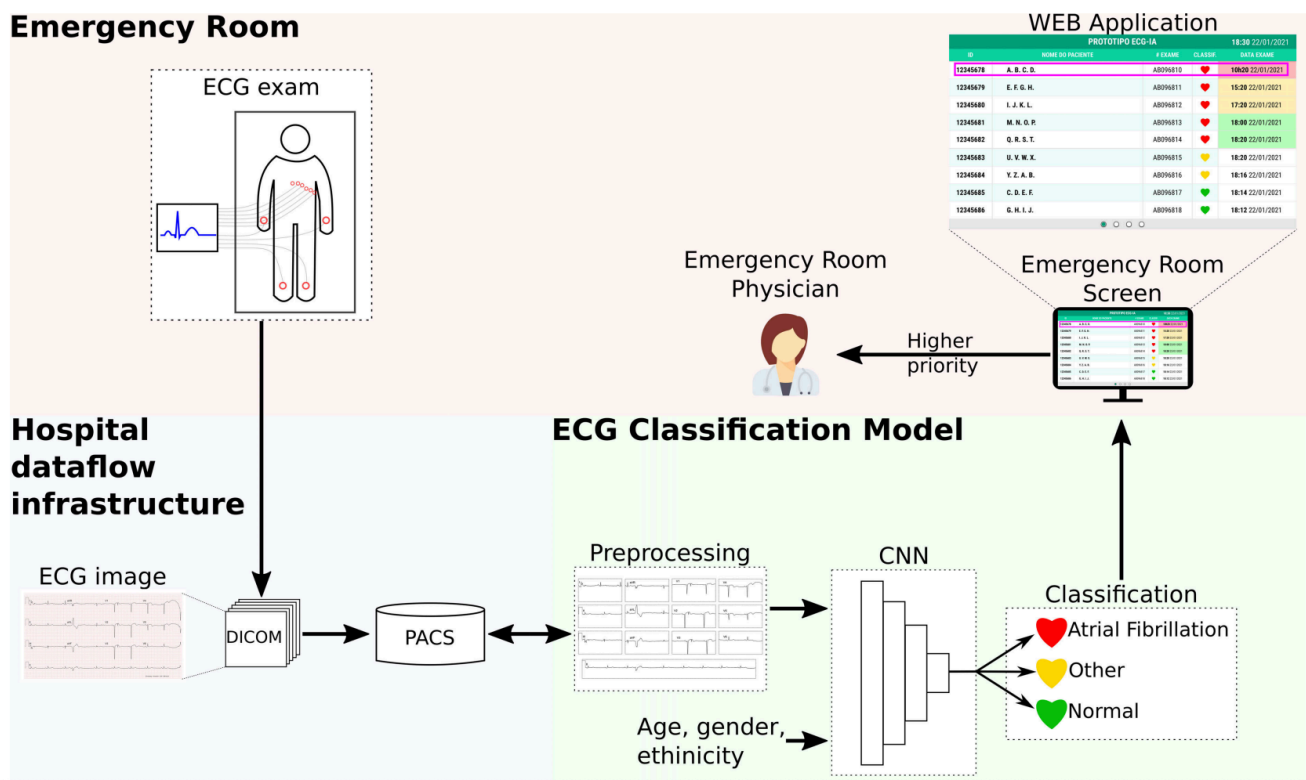


Figure 1 from Dias et al., IEEE Access (2023). Proposed integration of ECG acquisition, DICOM/PACS, classification and the emergency room web application. Image provided by the author.

The 12-lead examination enters the hospital infrastructure in DICOM format and is stored in PACS, the picture archiving and communication system. The AI module retrieves the ECG image, preprocesses it and runs a convolutional neural network. The diagram also includes age, sex and ethnicity as classifier inputs.

The model assigns one of three classes: atrial fibrillation, other or normal. A web application displays these classifications on the emergency room screen, using colors to help the clinical team identify examinations flagged for priority review.

Method development and first author of the IEEE Access paper

Team publications

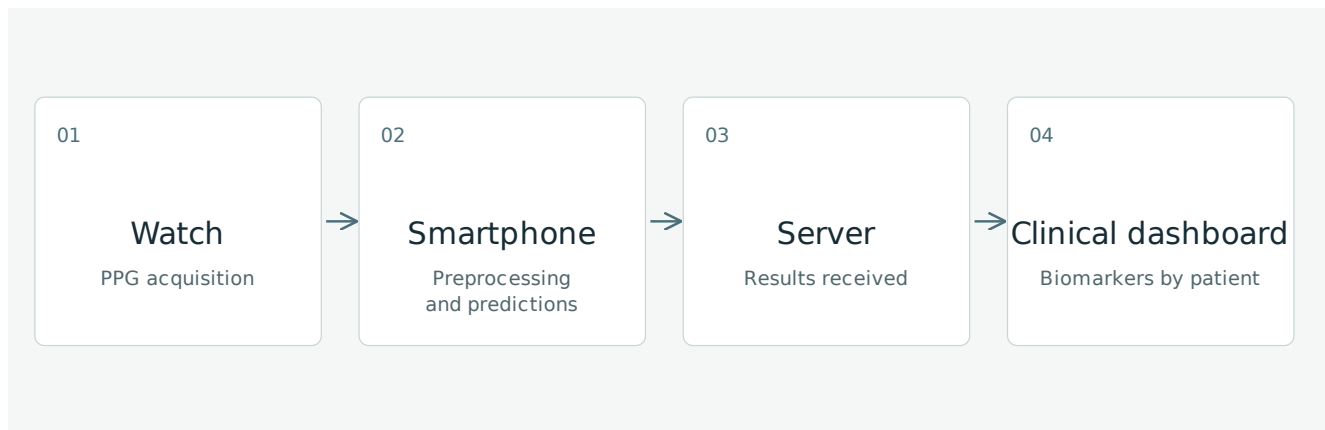
Artificial Intelligence-Driven Screening System for Rapid Image-Based Classification of 12-Lead ECG Exams: A Promising Solution for Emergency Room Prioritization. IEEE Access, 2023.

From wearable signals to clinical follow-up

Support patient follow-up outside the hospital by connecting wearable recordings to biomarkers available to the clinical team.

My role

My work covered collection protocols with prototype watches, data organization, photoplethysmography (PPG) preprocessing and collaborative algorithm development.



RPMS architecture. Predictions run on the smartphone.

The watch records PPG and sends the signals to a smartphone. The mobile app runs the predictions and transmits the results to the server. An online dashboard lets physicians review biomarkers and follow each patient over time.

Algorithms and research

Signal quality

Automated assessment of PPG segment quality.

Blood pressure

Calibrated estimation and evaluation of generalization limits.

Diabetes

Classification using PPG features and patient metadata.

Sleep

Sleep stage classification using temporal context.

Team publications

Quality Assessment of Photoplethysmography Signals For Cardiovascular Biomarkers Monitoring Using Wearable Devices. arXiv · Preprint, 2023.

Exploring the limitations of blood pressure estimation using the photoplethysmography signal. Physiological Measurement, 2025.

Machine Learning-Based Diabetes Detection Using Photoplethysmography Signal Features. SBCAS, 2024.

Enhancing Photoplethysmography-Based Sleep Staging Models Through Temporal Context Optimization. Studies in Health Technology and Informatics, 2025.